Pathfinder Software Requirements Specification (SRS) - REVISED

---

1. Scope

This SRS describes the software requirements for "Pathfinder," a web application developed using React, HTML, CSS, and Node.js. It is designed for high school and college students entering the workforce, offering comprehensive assistance through a personalized user interface. The core features include career guidance, job matching, resume analysis, and an AI-powered chatbot assistant.

---

2. Component Breakdown

2.1 Front-End Components

- **Landing Page:** Login/Sign-Up functionality with responsive design for various devices.
- **Dashboard:** Displays personalized user information and provides access to core features.
- **Chatbot UI:** AI chatbot accessible on all pages for quick assistance.
- **Profile Setup Page:** Allows users to enter and edit personal and career preference details.

2.2 Back-End Components

- **Authentication Service:** Handles user registration, login, and secure session management using Firebase.
- **Data Processing Service:** Processes user inputs such as career questionnaire responses and generates recommendations.
- **API Integration Service:** Interfaces with external job boards and OpenAI for real-time data and AI responses.

2.3 AI-Powered Chatbot Components

- **Query Processing:** Routes user queries to OpenAI API for response generation.
- **Context Management:** Retains session context for a conversational experience.

2.4 Database and Data Storage

- **Firebase Database:** Stores user profiles, preferences, and historical data securely.

2.5 Deployment

- Hosted on cloud platforms (e.g., AWS or Google Cloud) for scalability and reliability.

---

3. Requirements

3.1 Required States and Modes

- **Landing Page State:** Initial page for logging in or signing up.
- **Dashboard State:** Centralized hub displaying user-specific tools and information.
- **Settings State:** Allows customization of account settings and preferences.

3.2 Functional Requirements

- **FR1:** The website shall support user authentication via Firebase.
- **FR2:** The website shall provide personalized career suggestions based on a questionnaire.
- **FR3-1:** The website shall match jobs based on user-selected career paths.
- **FR3-2:** The job-matching system shall refresh results every 24 hours.
- **FR4:** The website shall analyze uploaded resumes and suggest improvements in specific areas.
- **FR5-1:** The website shall include an AI chatbot assistant for career-related guidance.
- **FR5-2:** The chatbot shall be accessible through a floating button in the bottom-right corner of all pages.
- **FR5-3:** The chatbot shall interact with users via a text-based interface with an optional voice input feature.
- **FR5-4:** Responses from the chatbot shall be generated via OpenAI API, with fallback responses for common queries.
- **FR5-5:** The chatbot shall offer direct links to additional resources for further guidance.
- **FR6:** Users shall be able to update career preferences and settings in their profile.
- **FR7:** The dashboard shall notify users of new job matches and updates to their career paths or resumes.

3.3 External Interface Requirements

- **EIR1:** Integrate with job board APIs for fetching relevant job listings.
- **EIR2:** Use Firebase for secure data storage and authentication.

3.4 Internal Data Requirements

- **IDR1:** User data, including profile and preferences, shall be securely stored in a Firebase database.
- **IDR2:** Questionnaire results and career path data shall be retrievable for real-time user interactions.

3.5 Adaptation Requirements

- **AR1:** The website shall adapt seamlessly to varying screen sizes, ensuring usability on mobile, tablet, and desktop devices.

3.6 Security Requirements

- **SecR1:** User data, such as email and passwords, shall be encrypted during transmission and storage.
- **SecR2:** Only authenticated users shall have access to personalized features.

3.7 Environment Requirements

- **ER1:** The application shall support major browsers, including Chrome, Firefox, Safari, and Edge.

3.8 Computer Resource Requirements

- **CRR1:** The system shall perform optimally on low-spec mobile devices.

3.9 Software Quality Factors

- **SQF1:** The interface shall be intuitive, minimizing the need for external user guidance.
- **SQF2:** Downtime for maintenance shall be communicated in advance to users.

3.10 Design and Implementation Constraints

- **DIC1:** The development process shall be constrained by the capabilities of React, Node.js, Firebase, and OpenAI APIs.

3.11 Personnel-related Requirements

- **PRR1:** The project team must include at least one full-stack developer proficient in the chosen tech stack.

3.12 Training-related Requirements

- **TRR1:** A built-in tutorial shall guide users through the application's features during the initial setup.

3.13 Other Requirements

- **OR1:** Accessibility features such as screen reader compatibility and keyboard navigation shall be implemented.

---

4. Precedence and Criticality of Requirements

Security and usability requirements are critical and shall take precedence in the development lifecycle.

---

5. Qualification Provisions

The application shall undergo comprehensive:

- **Usability Testing:** Evaluate user interface intuitiveness.
- **Performance Testing:** Assess response times and scalability.
- **Security Testing:** Verify data protection measures against breaches.

---

6. Requirements Traceability

A requirements traceability matrix shall map each requirement to its corresponding implementation and test case.

---

7. Notes

This document is subject to revisions based on iterative feedback from stakeholders and users.

---

8. Appendices

- **A. Glossary:** Contains technical terms used in this document.
- **B. Revision History:** Logs updates to this SRS.

---

9. Evaluation Criteria

The project's success will be determined by:

- Adherence to the specified requirements.
- Feedback from beta testers on usability and functionality.
- System performance under expected user loads.

---

10. Acronyms and Abbreviations

- **SRS:** Software Requirements Specification

- **UI:** User Interface
- **API:** Application Programming Interface